$$I = \int \frac{dx}{2+\cos x}$$

$$u = \tan\left(\frac{x}{2}\right)$$

$$dx = \frac{2du}{1+u^2}$$

$$\cancel{dx}\cos x = \frac{1-u^2}{1+u^2}$$

$$I = \int \frac{\frac{2du}{1+u^2}}{2+\frac{1-u^2}{1+u^2}} * \frac{1+u^2}{1+u^2}$$

$$I = \int \frac{2du}{2+2u^2+1-u^2}$$

$$I = \int \frac{2du}{u^2+3}$$

$$I = \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{u}{\sqrt{3}}\right) + C$$

$$I = \frac{3}{\sqrt{3}} \tan^{-1}\left(\frac{\tan\frac{x}{3}}{\sqrt{3}}\right) + C$$

$$\int \frac{dA}{\Delta^2 + a} = \frac{1}{\sqrt{a}} \tan^{-1}\left(\frac{\Delta}{\sqrt{a}}\right) + C$$

$$\int \frac{3}{2-3x}\,dx$$

$$= -\int \frac{-3}{2-3x}\,dx = -\text{len}(2-3x) + C$$

$$\int \frac{2}{1-5x}\,dx = \frac{-2}{5}\int \frac{-5}{1-5x}\,dx$$

$$= \frac{-2}{5}\,\text{len}(1-5x) + C$$

$$\int \frac{5}{3-5x}\,dx = \quad \text{let } u = 3-5x$$

$$\frac{du}{dx} = -5$$

$$du = -5\,dx$$

$$dx = -\frac{1}{5}\,du$$

$$\int \frac{5}{3-5x}\,dx = \frac{\cancel{5}}{u} \times -\frac{1}{\cancel{5}}\,du$$

$$= -\int \frac{1}{u}\,du$$

$$= -\text{len}(u) + C$$

$$= -\text{len}(3-5x) + C$$

$$\int \frac{1}{(x^2+1)(2x+1)}\, dx$$

$$= \int \frac{1}{(x^2+1)(2x+1)} = \frac{Ax+B}{(x^2+1)} + \frac{C}{2x+1}$$

$$1 = Ax + B(2x+1) + C(x^2+1)$$

$$1 = 2Ax^2 + (2B+A)x + B + Cx^2 + C$$

$$1 = (C+2A)x^2 + (2B+A)A + B + C$$

$$2A + C = 0 \qquad C = -2A$$

$$2B + A = 0 \qquad B = -\frac{A}{2}$$

$$B + C = 1$$

$$1 = -\frac{A}{2} - \frac{4}{2}A = \frac{-5A}{2} = 1$$

$$A = \frac{-2}{5} \;/\; B = \frac{1}{5} \;/\; C = \frac{4}{5}$$

$$\frac{1}{5}\int \frac{-2x+1}{x^2+1}\, dx + \frac{4}{5}\int \frac{1}{2x+1}\, dx$$

$$-\frac{1}{5}\ln(x^2+1) + \tan^{-1}(x) + C$$

Gözüm

mr / Ahmed aboaldahab

Integreation by Patil fraction

$$I = \int \frac{2x+1}{x^2+3x+2}\, dx$$

$$\frac{d}{dx} = (x^2+3x+2) = 2x+3$$

$$\frac{2x+1}{x^2+3x+2} = \frac{2x+1}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$$

$$= 2x+1 = A(x+2) + B(x+1)$$

$x = -1$, $A = -1$ | $x = -2$, $B = -3$

$$\boxed{A = -1} \qquad \boxed{B = -3}$$

$$\int \frac{-1}{x+1} + \frac{3}{x+2}$$

$$I = -\ln(x+1) + 3\ln(x+2) + C$$

GoodNotes

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$$\int \frac{3x+1}{5x^2+2}\,dx$$

$$\int \frac{3x}{5x^2+2} + \frac{1}{5x^2+2}\,dx$$

$$\frac{3}{10}\int \frac{10x}{5x^2+2} + \frac{1}{5}\int \frac{1}{x^2+2}\,dx$$

$$\frac{3}{10}\ln(5x^2+2) + \frac{1}{5}\cdot\frac{1}{\sqrt{\frac{2}{5}}}\tan^{-1}\left(\frac{x}{\sqrt{\frac{2}{5}}}\right)$$

$$\int \frac{\sin x \cos x \tan x}{\sin \cos \tan x}$$

$$\boxed{u = \tan\left(\frac{x}{2}\right)}$$

$$du = \frac{1}{2}\sec^2\left(\frac{x}{2}\right)dx$$

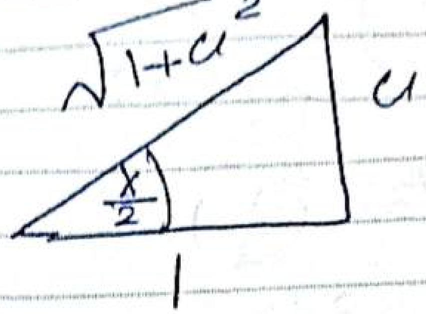
$$2du = \sec^2\left(\frac{x}{2}\right)dx$$

$$2du = \left[1+\tan^2\left(\frac{x}{2}\right)\right]dx$$

$$2du = (1+u^2)\,dx$$

$$\boxed{dx = \frac{2du}{1+u^2}}$$

$\sin x$



$$\sin x = 2\sin\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right)$$

$$= 2\left(\frac{u}{\sqrt{1+u^2}}\right)\left(\frac{1}{\sqrt{1+u^2}}\right)$$

$$\boxed{\sin(x) = \frac{2u}{1+u^2}}$$

$$\cos x = \cos^2\frac{x}{2} - \sin^2\frac{x}{2}$$

$$= \frac{1}{1+u^2} - \frac{u^2}{1+u^2}$$

$$\boxed{\cos x = \frac{1-u^2}{1+u^2}}$$

$$\cos^2 x + \sin^2 x = 1$$
$$1 + \tan^2 x = \sec^2 x$$
$$\cot^2 x + 1 = \csc^2 x$$
$$\sin 2x = 2\sin x \cos x$$
$$\cos 2x = \cos^2 x - \sin^2 x$$

حفظ

$$\int \frac{x^3 - 2x}{x^2 + 3x + 2}\, dx$$

$$\begin{array}{r|l} & x - 3 \\ \hline x^2 + 3x - 2 & x^3 - 2x \\ & -x^3 + 3x^2 + 2x \\ \hline & -3x^2 - 4x \\ & -3x^2 - 9x + 6 \\ \hline & 5x - 6 \end{array}$$

$$\int \frac{x^3 - 2x}{x^2 - 3x + 2}\, dx = \int x - 3 + \frac{5x - 6}{x^2 + 3x + 2}$$

$$= \frac{1}{2}x^2 - 3x \int \frac{5x - 6}{(x+1)(x+2)}$$

$$\int \frac{5x - 6}{(x+1)(x+2)} = \frac{A}{(x+1)} + \frac{B}{(x+2)}$$

$$5x - 6 = A(x - 2) + B(x + 1)$$

$x = -2 \qquad B = 16$

$x = -1 \qquad A = -11$

$$= \frac{1}{2}x^2 - 3x - 11 \ln(x+1) + 16 \ln(x+2)$$

Goal